Measure voltage between two points

Task 4.5 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Label node pairs and lead colors. Place the DC voltmeter across each selected pair. Record source voltage, lamp drop and the sign when leads are reversed.

Record the source voltage across its terminals. Label measurement points on a schematic.

Pair of nodes	Red lead at	Black lead at	Prediction and unit
Source			
Lamp			
Closed connection			

02 Plan it on paper

Predict the voltage across a lamp and across a closed connection. Mark both meter leads for each prediction.

My prediction, chosen feature or design limit

Measure voltage between two points

Task 4.5 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Connect the voltmeter in parallel at each pair of points. Record the setting, sign, and units. Reverse the leads on one measurement. Explain the sign change and compare the voltage drops round a closed loop.

Pair / condition	DC range	Signed reading (V)	Explanation
Source			
Lamp			
Connection			
Reversed pair			

04 Use the reading

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Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Show one lead placement and explain what the number measures between those two points.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.